

# PHYS 3717: Particle Physics

<https://homiller-hep.net/teaching/2026-2/>

## Homework 3

Deadline: September 29, 2026 (in class)

### Problem 1: Lie Algebra Identities

The Lie algebra with generators  $t_a$  and structure constants  $C^a_{bc}$  is written as

$$[t_b, t_c] = iC^a_{bc}t_a. \quad (1)$$

- (a) Show that the structure constants satisfy the constraint

$$C^a_{bc}C^e_{ad} + C^a_{cd}C^e_{ab} + C^a_{db}C^e_{ac} = 0. \quad (2)$$

As a hint: take the commutator of Eq. (1) with the generator  $t_d$ , then add to this the same expression with the indices  $b, c, d$  permuted to  $c, d, b$  and  $d, b, c$ . Then argue that the left-hand side of the resulting expression vanishes identically. This is known as the *Jacobi identity*.

- (b) Show that the matrices

$$(T_a)^b_c = -iC^b_{ca} \quad (3)$$

satisfy the commutation relations in Eq. (1), and thus form a representation of the Lie algebra. This is known as the *adjoint representation*.

### Problem 2: The Poincaré Algebra

In this problem we'll work out a few of the intermediate steps in the derivation of the Poincaré transformations of  $J^{\rho\sigma}$  and  $P^\rho$ .

- (a) Using the general operator composition rule  $U(\bar{\Lambda}, \bar{a})U(\Lambda, a) = U(\bar{\Lambda}\Lambda, \bar{\Lambda}a + \bar{a})$ , show that

$$U(\Lambda, a)U(1 + \omega, \epsilon)U^{-1}(\Lambda, a) = U(\Lambda(1 + \omega)\Lambda^{-1}, \Lambda\epsilon - \Lambda\omega\Lambda^{-1}a) \quad (4)$$

- (b) Starting from the following relation derived in the notes,

$$U(\Lambda, a)\left[-\frac{1}{2}\omega_{\rho\sigma}J^{\rho\sigma} - \epsilon_\rho P^\rho\right]U^{-1}(\Lambda, a) = -\frac{1}{2}(\Lambda\omega\Lambda^{-1})_{\mu\nu}J^{\mu\nu} - (\Lambda\epsilon - \Lambda\omega\Lambda^{-1}a)_\mu P^\mu, \quad (5)$$

show that

$$U(\Lambda, a)J^{\rho\sigma}U^{-1}(\Lambda, a) = \Lambda_\mu{}^\rho\Lambda_\nu{}^\sigma[J^{\mu\nu} - a^\nu P^\mu + a^\mu P^\nu]. \quad (6)$$

- (c) Starting from the equation derived in part (b), derive the following relation for an infinitesimal Poincaré transformation defined by  $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$  and  $a^\mu = \epsilon^\mu$ , with  $\omega$  and  $\epsilon$  infinitesimal:

$$-i\left[\frac{1}{2}\omega_{\mu\nu}J^{\mu\nu} + \epsilon_\mu P^\mu, J^{\rho\sigma}\right] = \omega_\mu{}^\rho J^{\mu\sigma} + \omega_\mu{}^\sigma J^{\rho\mu} - \epsilon^\sigma P^\rho + \epsilon^\rho P^\sigma. \quad (7)$$

### Problem 3: SU(2) Subalgebras of the Poincaré Algebra

In class, we derived the Lie algebra for the generators  $J^{\mu\nu}$  and  $P^\mu$  of Poincaré transformations. The generators of Lorentz transformations in particular, satisfy:

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(g^{\nu\rho} J^{\mu\sigma} - g^{\mu\rho} J^{\nu\sigma} - g^{\sigma\mu} J^{\rho\nu} + g^{\sigma\nu} J^{\rho\mu}) \quad (8)$$

(a) Define

$$J^i \equiv \frac{1}{2} \epsilon^{ijk} J^{jk}, \quad K^i \equiv J^{0i}, \quad (9)$$

for  $i, j, k = 1, 2, 3$ . These are the generators of rotations and boosts, respectively. From Eq. (8), show that these satisfy

$$[J^i, J^k] = i\epsilon^{ijk} J^k, \quad [J^i, K^j] = i\epsilon^{ijk} K^k, \quad [K^i, K^j] = -i\epsilon^{ijk} J^k. \quad (10)$$

We see that the rotation generators (angular momentum operators) satisfy the commutation relations of a closed SU(2) Lie algebra, while the boost generators have mixed commutation relations with the  $J^i$ .

(b) Now, define the linear combinations

$$\vec{A} \equiv \frac{1}{2}(\vec{J} + i\vec{K}), \quad \vec{B} \equiv \frac{1}{2}(\vec{J} - i\vec{K}). \quad (11)$$

Show that with these linear combinations, the algebras decouple and the  $\vec{A}$  and  $\vec{B}$  satisfy

$$[A^i, A^j] = i\epsilon^{ijk} A^k, \quad [B^i, B^j] = i\epsilon^{ijk} B^k, \quad [A^i, B^j] = 0. \quad (12)$$

This fact can (and will!) be used to identify the finite-dimensional representations of the Lorentz group later on.<sup>1</sup>

### Problem 4: Explicit Forms of Lorentz Generators

(a) Show that

$$(\mathcal{J}^{\mu\nu})^\alpha{}_\beta \equiv i(g^{\mu\alpha} \delta^\nu_\beta - g^{\nu\alpha} \delta^\mu_\beta) \quad (13)$$

satisfies the commutation relations in Eq. (8) above, and thus provides an explicit representation of the generators of Lorentz transformations. Note that the generators have *labels* that are spacetime indices,  $\mu, \nu$ , and that for each choice of  $\mu, \nu$ , one can regard  $\mathcal{J}^{\mu\nu}$  as a  $4 \times 4$  matrix with row index  $\alpha$  and column index  $\beta$ .

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<sup>1</sup>Note, however, that  $A$  and  $B$  are not Hermitian—in fact,  $(A^i)^\dagger = B^i$  and vice-versa. Thus while this algebraic relation will be useful in identifying representations of the Lorentz group, these operator combinations do not correspond to any physical observables.

- (b) Find the explicit form for the matrix  $\mathcal{K}^1 = \mathcal{J}^{10}$ . Consider  $\omega_{10} = \phi \neq 0$ , while all other  $\omega_{\mu\nu} = 0$ . Compute the explicit form of the finite Lorentz transformation matrix

$$\Lambda^\alpha{}_\beta = \exp\left(-\frac{i}{2}\omega_{\mu\nu}\mathcal{J}^{\mu\nu}\right)^\alpha{}_\beta. \quad (14)$$

Show that this corresponds to a boost along the  $x$ -direction, and relate  $\phi$  to the usual expressions for the Lorentz boost factors  $v$  and  $\gamma = (1 - v^2)^{-1/2}$  (with  $c = 1$ ).